# Master of Science in Computer Science
# MSc Thesis Registration Form

**Preliminary Title of the Thesis**

Resilience and Critical Failure Points of Federated Learning: A Stress-Test on Data Heterogeneity and Privacy Noise

**Abstract**

*The collaborative training of AI models in healthcare is often impacted by data isolation and strict privacy regulations, such as GDPR and HIPAA. Federated Learning (FL) is a solution that addresses these challenges by allowing institutions to train models locally without sharing raw data. However, the interaction between data heterogeneity (non-IID) and privacy-preserving techniques has the potential to lead to a significant reduction in model utility. This thesis investigates the operational limits of a secure FL framework using public datasets. The objective is to determine the 'failure points' where a federated system ceases to provide clinical utility compared to localized or centralized models.*

**Thesis Project Description**

*Federated Learning (FL) was introduced to mitigate the privacy risks associated with centralised data aggregation, by enabling institutions to train machine learning models on locally distributed datasets. Under this paradigm, individual sites participate in collaborative training by exchanging only model parameters (weights or gradients) with a central aggregator, rather than sharing raw patient records.*
*However, despite these natural privacy-preserving capabilities, research indicates that the framework remains susceptible to complex breaches, specifically Inference and Reconstruction attacks. In these scenarios, malicious actors can exploit model updates to extract sensitive patient information, proving that keeping data local does not always guarantee complete confidentiality.*

*To address these challenges and ensure GDPR/HIPAA compliance, it is essential to implement privacy-preserving techniques such as Differential Privacy (DP). However, these techniques introduce a critical trade-off: injecting noise to preserve privacy often degrades the model's accuracy and diagnostic usability. Additionally, healthcare data is heterogeneous across different hospitals due to varying patient populations and clinical specializations. This non-IID nature has the potential to result in "client drift," which can further compromise model performance.*

*The idea of this thesis is to develop and analyze a privacy-preserving FL framework that simulates real-world clinical scenarios characterized by heterogeneous data (Non-IID) and analyze the synergy*

*between data imbalance and privacy noise. The goal is to identify the 'failure points' where the global model's diagnostic value is inferior to centralized benchmarks or even simple localized training.*

*Objectives*

1. **Combined Sensitivity Testing:** conducted by varying the interaction between data heterogeneity (simulated via the Dirichlet Concentration Parameter $\alpha$) and privacy-preserving noise (via Differential Privacy Budget $\epsilon$). The objective is to perform a combined simulation across a range of $\alpha$ and $\epsilon$ values to map the model's resilience and behavioural boundaries under combined stress.

2. **Finding the System's Breaking Points:** the objective is to determine the specific thresholds where a federated model fails to provide clinical utility. This occurs when extreme class imbalance or excessive privacy noise leads to the suppression of diagnostic signals for under-represented diseases, making collaborative aggregation less effective than local training. Log-loss will be utilized alongside standard metrics to detect model instability, as it specifically penalizes confident but incorrect predictions.

3. **Clinical Utility Benchmarking:** Compare the federated framework against a centralized and the average local to justify the benefit of the collaborative approach.

*Expected Results and Products*

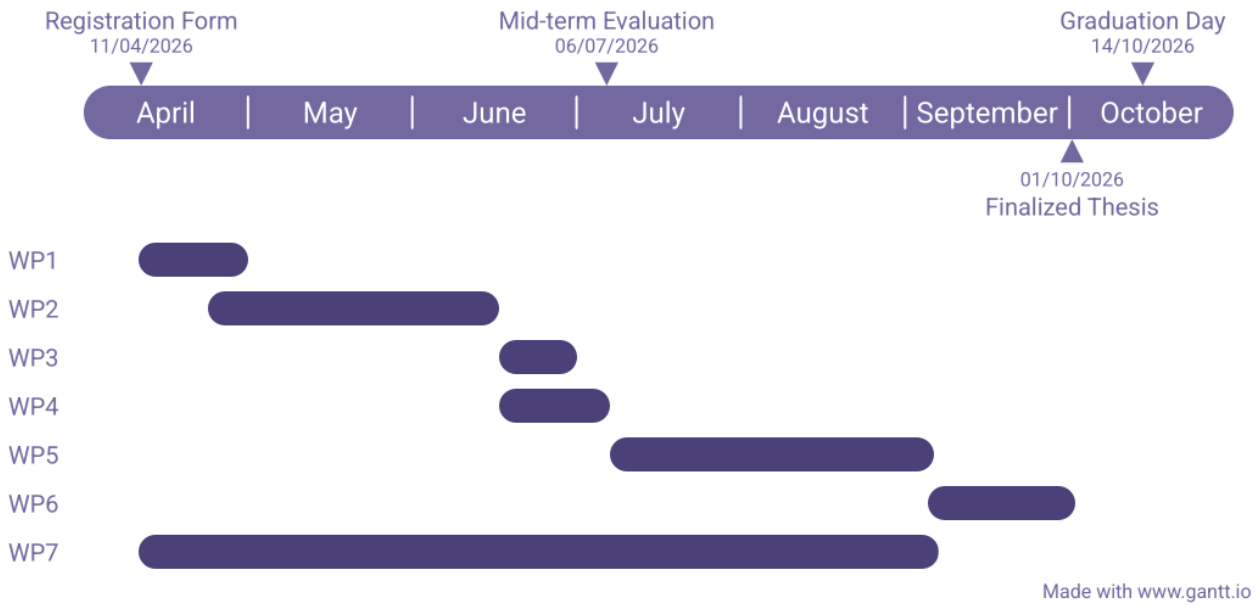
- A visual report containing the interaction between data heterogeneity ($\alpha$) and privacy budgets ($\epsilon$) to easily understand the behavioral boundaries.
- A series of thresholds that identify the critical points where global model performance degrades and federated aggregation loses its clinical utility.
- A probabilistic reliability report created by analyzing Log-loss to detect "hidden" failure points. In these cases, the model remains accurate but provides unreliable confidence levels for medical diagnoses.
- A table to compare federated, centralized, and local models results to justify the collaborative approach and quantify the "cost of privacy".
- A FL framework integrating FedProx and Differential Privacy to simulate and test the global model resilience in heterogeneous healthcare environments

*Workpackages and Tasks*

- WP1: Literature Review and Environment Setup. Researching Federated Learning, Differential Privacy, and aggregation methods, while selecting the dataset and researching useful tools and libraries.

- WP2: Framework Development. Implementing the privacy-preserving FL framework for non-IID data alongside centralized and local models.
- WP3: Initial experimental phase. Executing preliminary simulations to validate system stability.
- WP4: Mid-term Evaluation. Preparing the documentation and presentation required for the July progress assessment.
- WP5: Testing and Benchmarking. Conducting complete tests to identify "failure points" and comparing results against centralized and average local models.
- WP6: Thesis Finalization. Final editing, refining the visual reports, and completing the definitive thesis.
- WP7: Continuous Thesis Drafting. Iterative writing of the thesis chapters concurrently with WP1-WP5.

*Project Plan and Gantt Diagram*




**Student (or Students in case of Group Thesis)**

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*Reykjavik University (Iceland)*

**Intended Final Exam Session**

*Autumn 2025/2026 – Graduation Day 14th October 2026*

**Date of Signature**

____*10/04/2026*_____

**Student Signature**

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(repeat for each student if Group Thesis)

**Supervisor Signature**

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**Attachments and Requisites**

The student (students if group thesis) attaches to this document a screenshot of the student's career (or of the students' careers in case of group thesis), resulting from the student career system ESSE3, certifying that he/she have already acquired 78 ECTS credits (tolerance of 12 ECTS credits with respect to the 90 that must be acquired with exams to get the MSc).

Moreover, in the period from the date of signing of this Thesis Registration Form to the graduation day the student(s) must not acquire more than 12 ECTS credits (possibly after changes to the study plan).

**Notes**

This Thesis Registration Form (including the attachments), filled in and signed, must be sent by email to the Coordinator of the MSc in Computer Science, in cc the supervisors and all the co-supervisors, using the official email address of the student name.surname@studenti.unicam.it

The deadline for sending this email is the Friday of the 22nd week preceding the one in which the graduation day of the selected final exam session is scheduled. August is considered to have zero weeks in this computation. If the resulting Friday is a holiday then the deadline is postponed to the first following working day.

For the scheduled final exam sessions and the relative deadlines, consult

http://didattica.cs.unicam.it/doku.php?id=informazioni:thesisprocedures

This document can be downloaded from the same page.

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